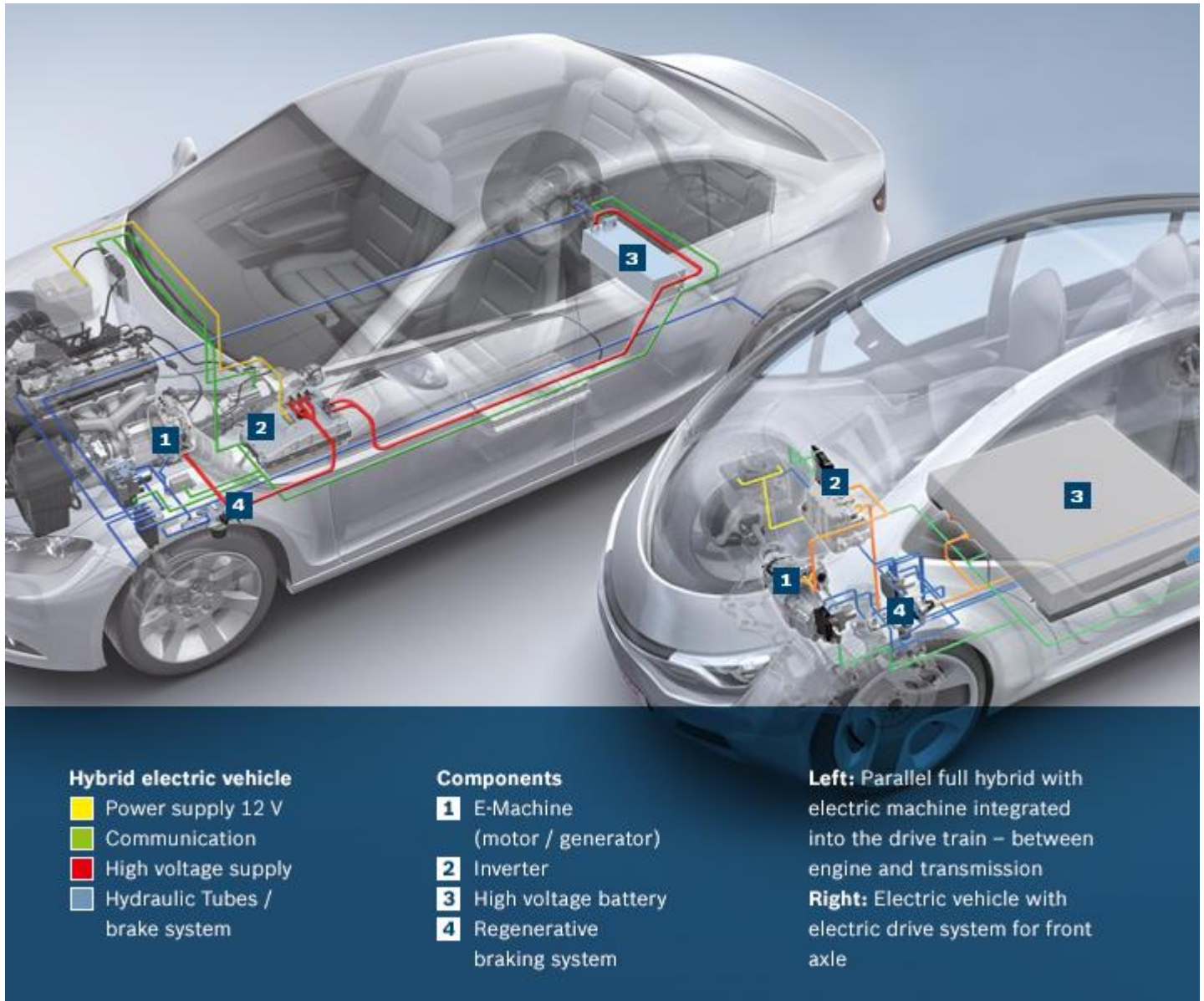
Colorado State University

Parallel Hybrid Design and Intro to Optimization



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Thomas Bradley
Associate Professor
Mechanical Engineering

Fall 2017

Parallel Hybrid Design and Intro to Optimization

Parallel Architecture Component Sizing Summary Optimization

- Review the overall operation of a parallel architecture HEV
  - Advantages
  - Disadvantages
- How to size the powertrain components for the course project
  - Example
  - Internal combustion engine
  - Electric motor
  - Transmission
  - Battery
- Optimization
  - Introduction

Parallel Hybrid Design and Intro to Optimization

Parallel Architecture

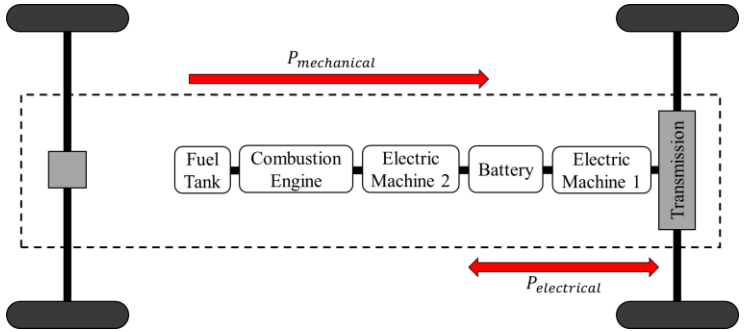
Component Sizing

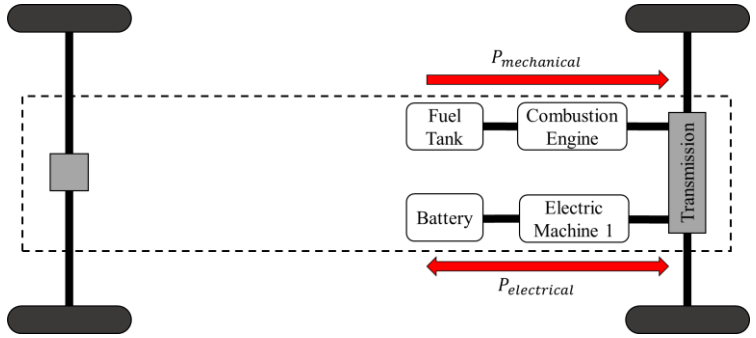
Summary

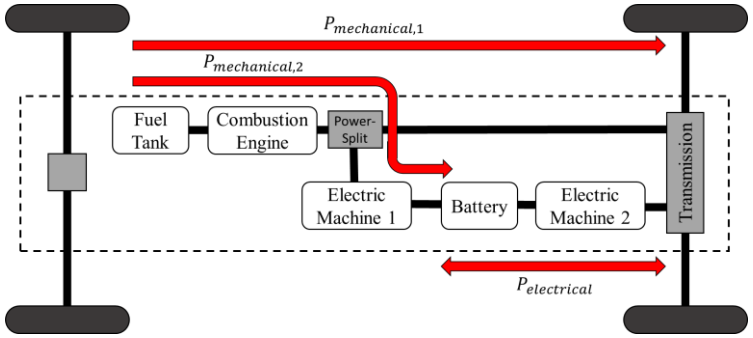
Optimization

<u>Don't rush to get your project done!</u>

- ✓ Series
- Parallel
- Series/Parallel or Power-Split (Toyota Prius)



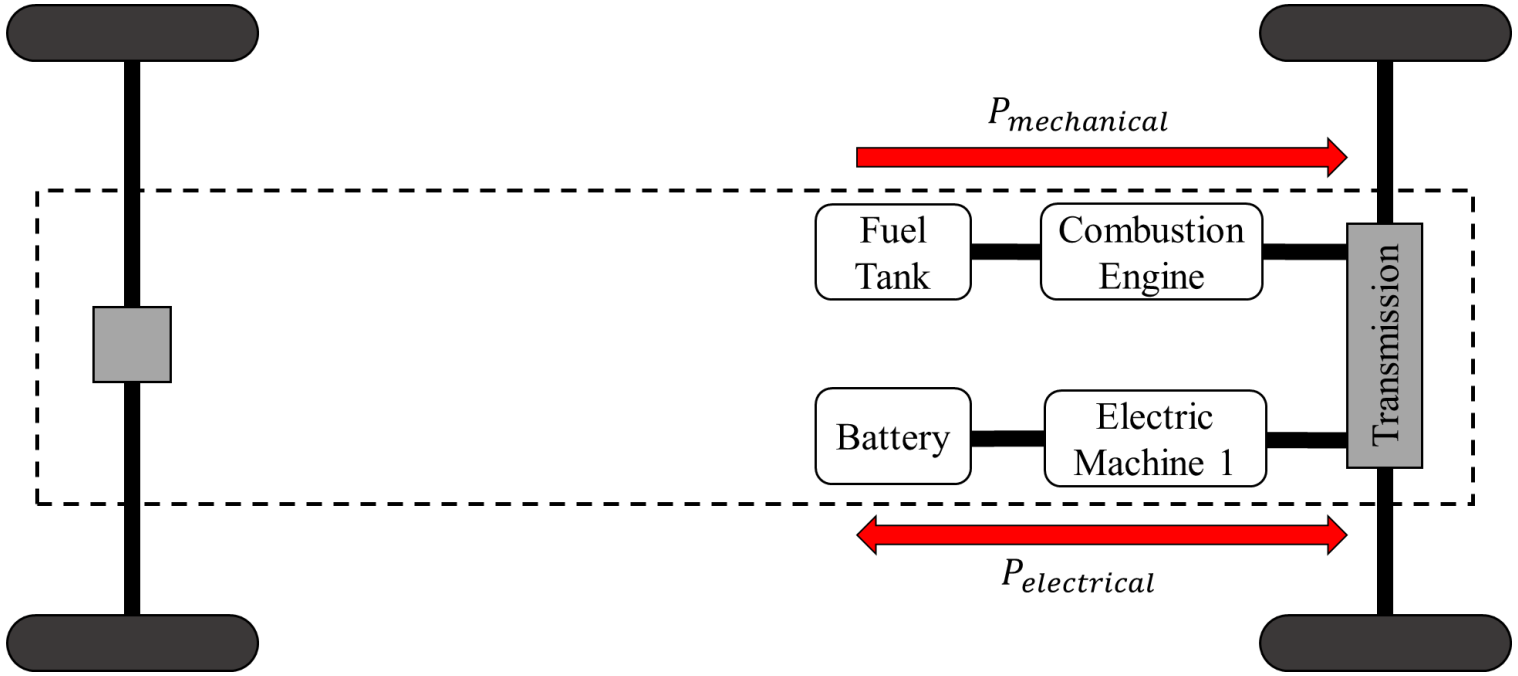




Parallel Hybrid Design and Intro to Optimization

Parallel Architecture Component Sizing Summary Optimization

- Recall the basics of parallel HEV operation…



Parallel Hybrid Design and Intro to Optimization

- Recall the basics of parallel HEV operation…

<u>Parallel Hybrid Advantages</u>

- Electric motor can double as a generator
  - No generator required
- Smaller components
- Fewer energy conversion efficiency penalties
- More energy captured during regenerative braking

<u>Parallel Hybrid Disadvantages</u>

- Control strategies governing two power sources are more complex
- Drivetrain components are more complex
- Larger engine required
- Engine cannot always operate at peak efficiency
  - Engine speed coupled to wheel speed

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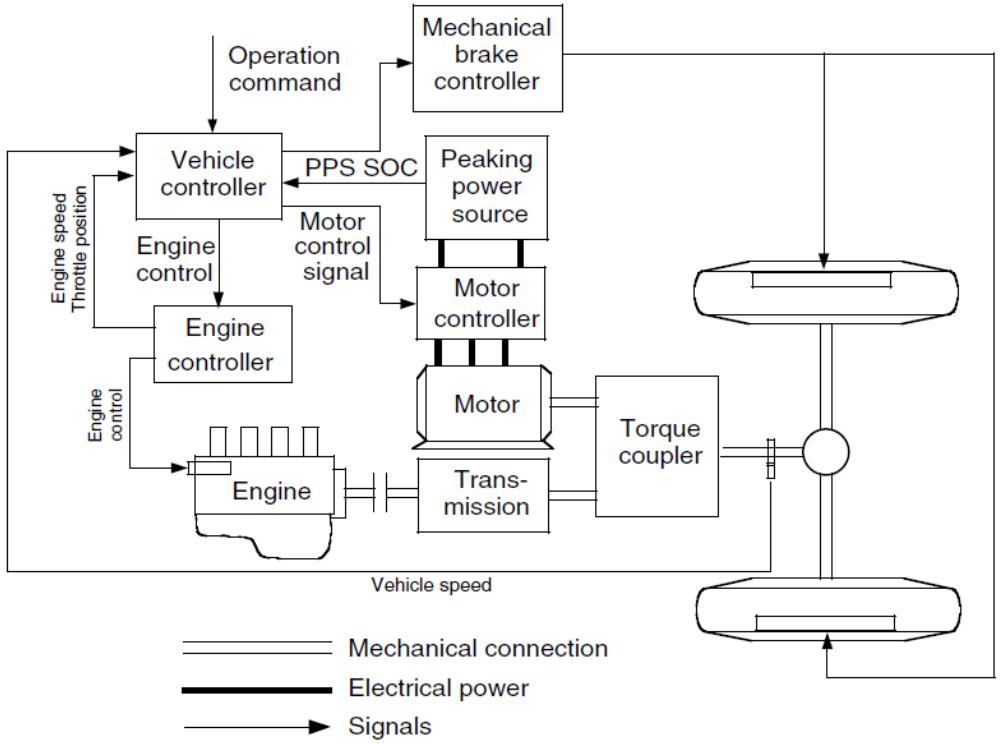
- Lets think about modeling this architecture…



Parallel Hybrid Design and Intro to Optimization

- Example: Parallelizing a conventional vehicle
  - Honda does this: 2015 Honda Civic
    - 1.5L V-TEC SOHC 4 Stroke Engine
    - Integrated Motor Assist
    - 1280 kg Curb Weight
  - Similar to the conventional version

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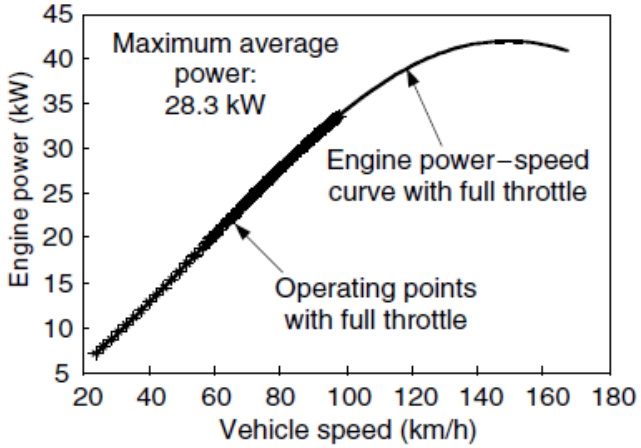
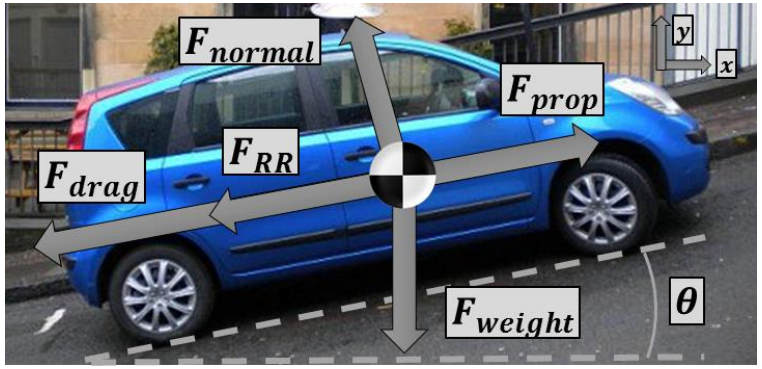
Parallel Architecture Component Sizing Summary Optimization

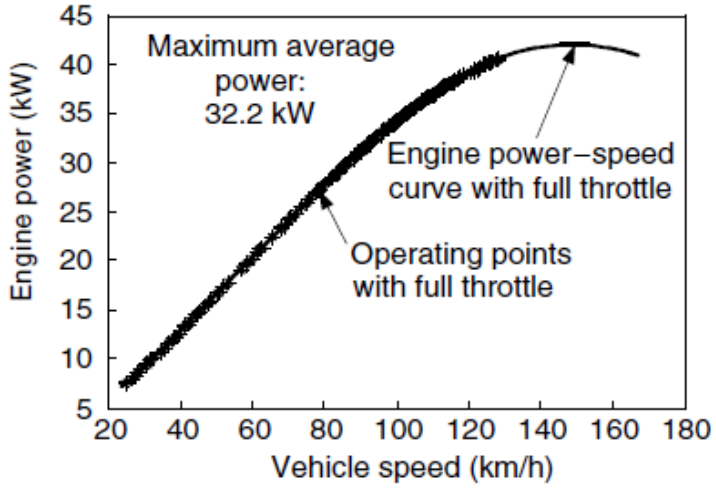
- Internal combustion engine
  - Larger than for series architecture
  - Support vehicle at normal constant speeds
  - But, engine speed = *f* (vehicle speed)
    - ➢ Not as easy to ensure proper engine operation
  - Choose initial engine size, evaluate performance



(b) FTP 75 highway driving cycle



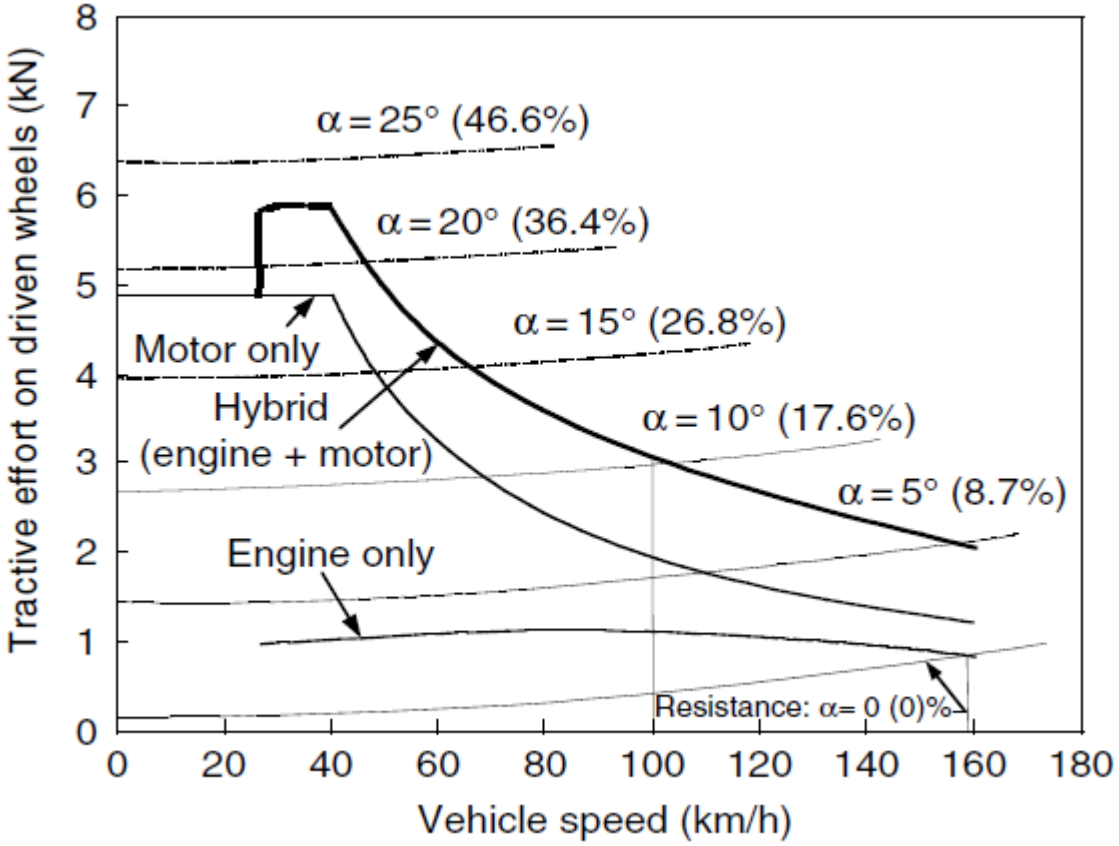
$$P_e = \frac{V}{1000\eta_{t,e}}\left(M_v g f_r + \frac{1}{2}\rho_a C_D A_f V^2 + M_v g\, i\right)(\text{kW})$$



(c) US06 driving cycle

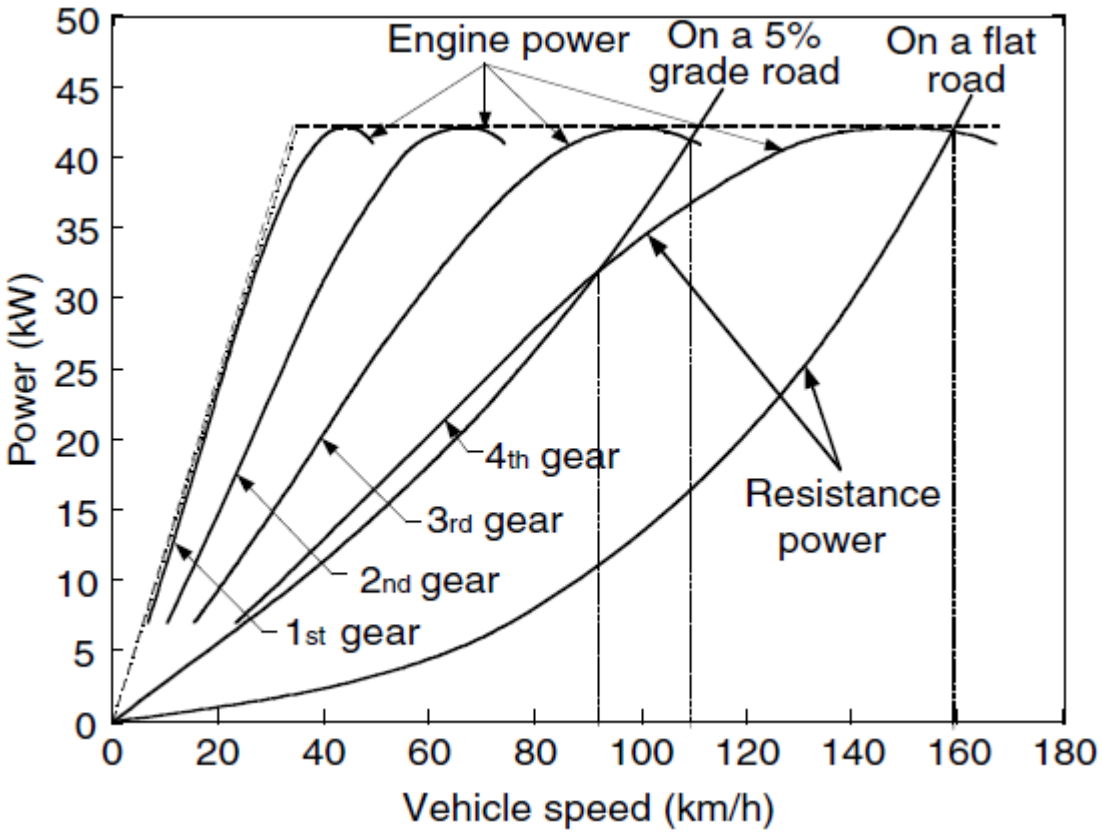
Parallel Hybrid Design and Intro to Optimization

- Electric motor
  - Can assist in during high power demand
  - Provide power during accelerations
    - Engine power for drag and rolling resistance
    - EM power during accelerations
    - Run simulations



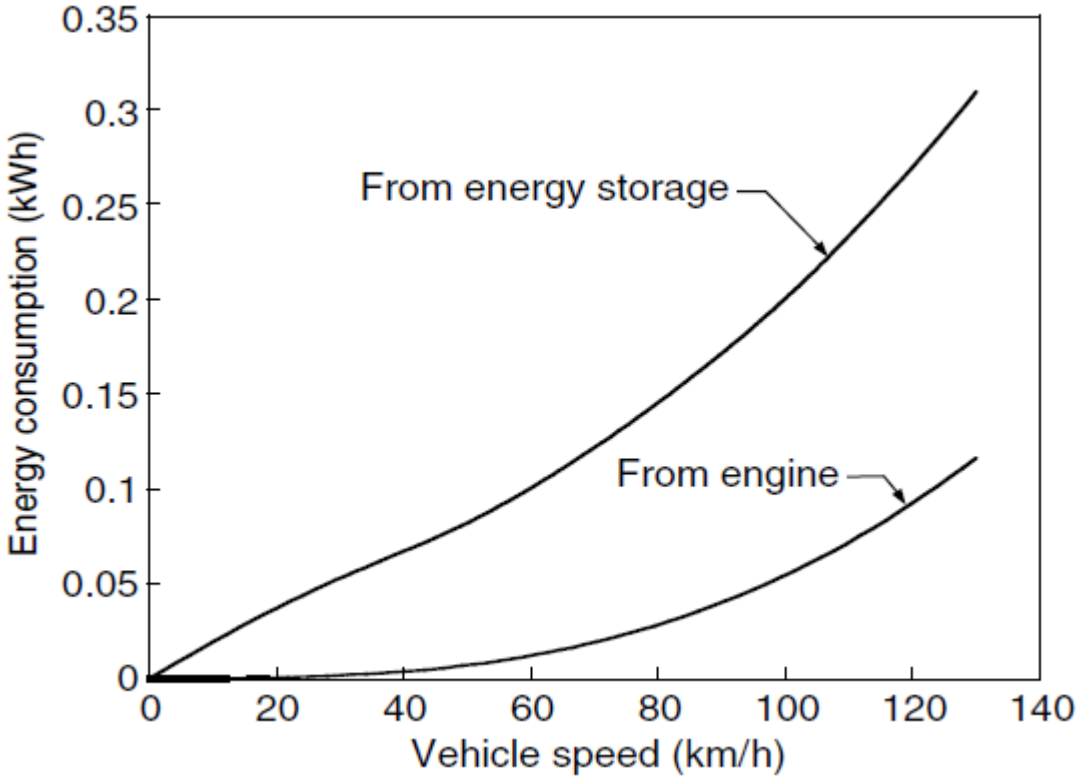
Parallel Hybrid Design and Intro to Optimization

- Transmission
  - 1st gear
    - Ensure hill climbing is possible
    - Ensure acceleration power demands are possible
  - Other gears
    - Higher overall engine efficiency
    - Improved vehicle performance
    - Improved battery charging
    - Improved fuel economy
    - But: Heavier and more difficult to control



Parallel Hybrid Design and Intro to Optimization

- Battery size
  - Meet high power demands
  - Assist with stop and go driving
  - Typically want to keep the state of charge high



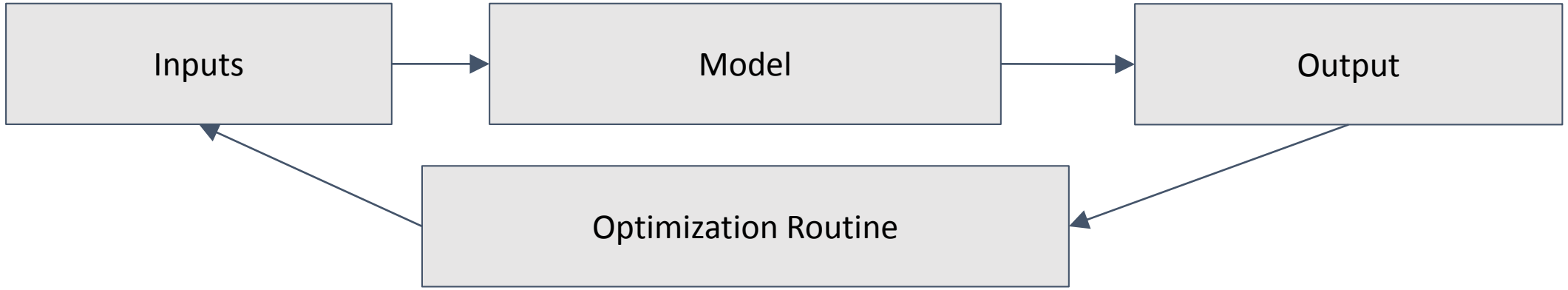
Parallel Hybrid Design and Intro to Optimization

- Parallel HEVs
  - Can significantly improve a conventional vehicle with a few modifications
  - The control strategies can be difficult
  - Engine operation is not as efficient
  - Balance engine, EM, battery, and transmission
    - Driving power demands are met
    - Engine is sized down from a conventional vehicle
    - EM assists with high power demand
    - EM provides power during stop and go
      - Fuel consumption reduction

Parallel Hybrid Design and Intro to Optimization

- Why do we care about optimization?

Definition of **optimization**. : an act, process, or methodology of making something (as a design, system, or decision) as fully perfect, functional, or effective as possible; specifically : the mathematical procedures (as finding the maximum of a function) involved in this.



Parallel Hybrid Design and Intro to Optimization

- Optimization is an entire discipline
- Here is a list of subjects taught in a generic optimization course:

<u>Unconstrained Optimization</u>

- 1-D Search Methods
  - Golden Search Method
  - Fibonacci Search Method
  - Newton's Method
  - Secant Method
- 2-D Search Methods
  - Steepest Descent
  - Conjugate Gradient
  - 2-D Newton's Method
- 3-D Search Methods
  - Quasi Newton: DFP algorithm
  - Quasi Newton: BFGS algorithm

<u>Constrained Optimization</u>

- Linear Programming
  - Graphical Solutions
  - Simplex Method
- Quadratic Programming
  - Lemke's algorithm
  - KKT conditions
- Dynamic Programming
- 2nd Order Cone Programming

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Parallel Hybrid Design and Intro to Optimization

Parallel Architecture Component Sizing Summary Optimization

- Optimization is an entire discipline → *But its actually not too scary…*

Parallel Hybrid Design and Intro to Optimization

- Example 1

Maximize: $Z = 6x_1 - 3x_2$

Subject to: $-\frac{2}{5}x_1 > 2$

$-\frac{3}{2}x_1 > 20$

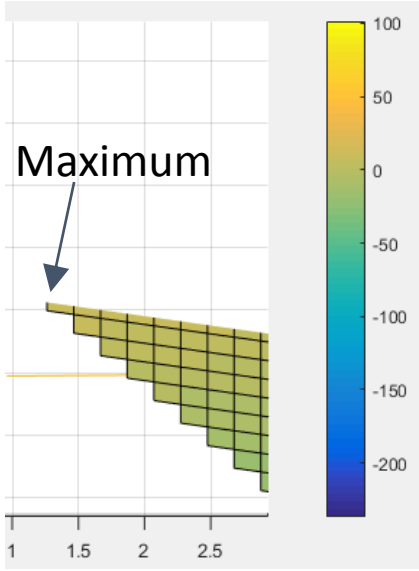
```
clear all; close all; clc;

%Create numerical variables
disc=100;
x1=linspace(-35,15,disc);        x2=linspace(-5,15,disc
X1=repmat(x1,disc,1);            X2=repmat(x2',1,disc);

%Define objective function and constraints
Z=6.*X1-3.*X2;
c1=-(2/5)*x1+2;
c2=-(3/2)*x1+20;

%Remove solutions that violate constraints
[~,I_c1]=min(abs(bsxfun(@minus,(-(2/5)*x1+2),x2')));
[~,I_c2]=min(abs(bsxfun(@minus,-(3/2)*x1+20,x2')));
for i=1:disc
        Z(1:I_c1(i),i)=nan;
end
for i=1:disc
    if I_c2(i)==disc
        continue;
    else
        Z(I_c2(i)+1:disc,i)=nan;
    end
end

%PLot Results
figure(2)
xlim([-35 15]);     ylim([-5 15]);
hold on
plot([-35,-5,15],[15,15,15]);   plot([15,15,15],[-5,5,15]);
plot(x1,c1);                    plot(x1,c2);
surf(X1,X2,Z);
```

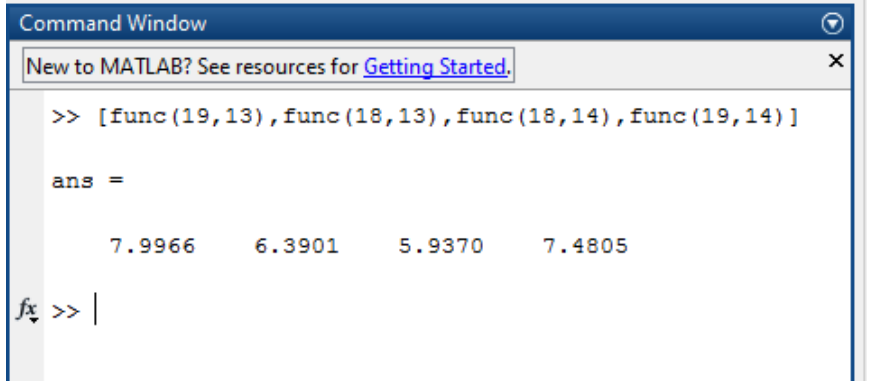


Parallel Hybrid Design and Intro to Optimization

- Example 2

Parallel Hybrid Design and Intro to Optimization

- Example 2: Say we have some dynamic model...



How do we find the minimum?

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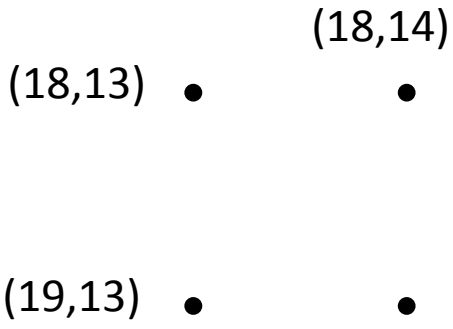

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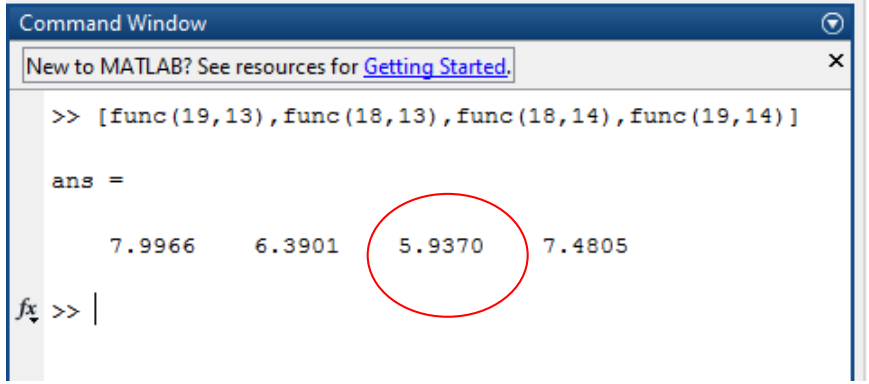

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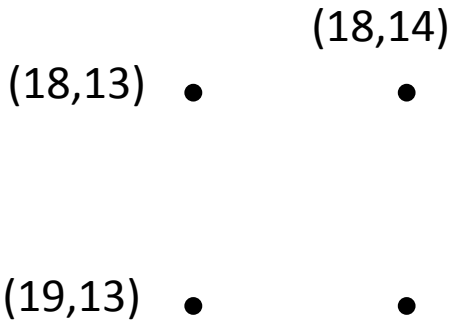
2. Find the minimum (check boundary points)



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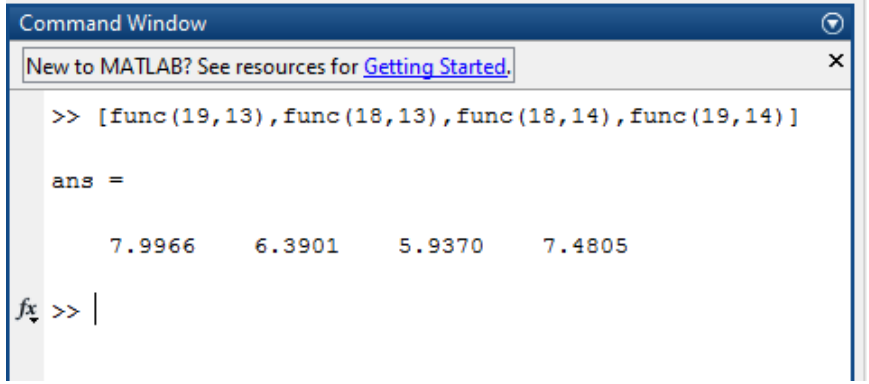

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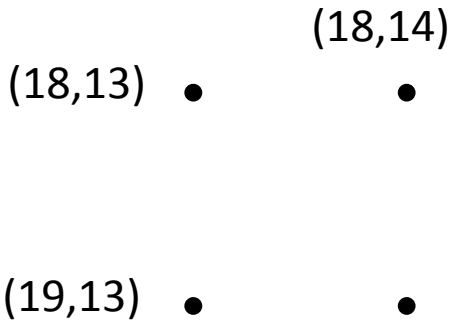

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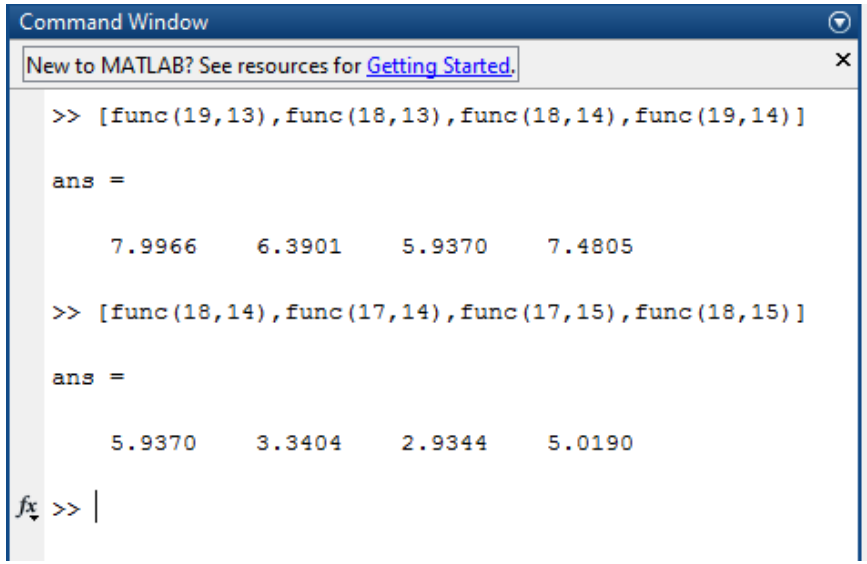

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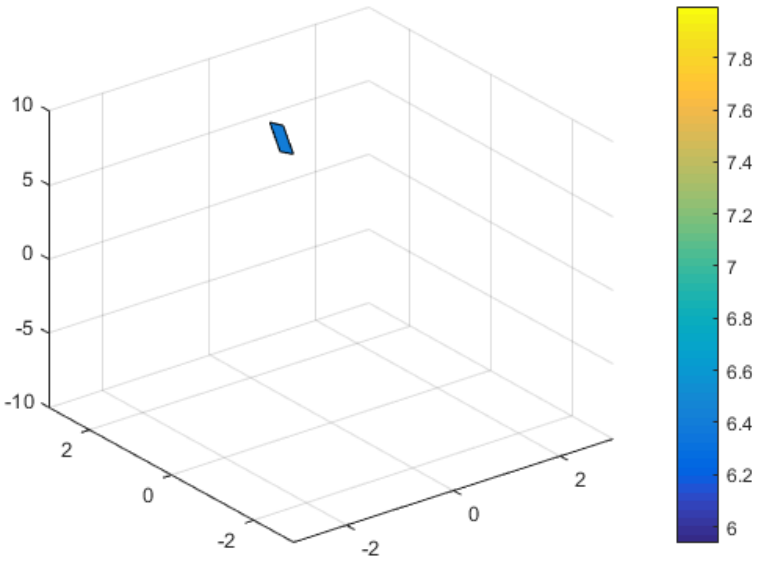

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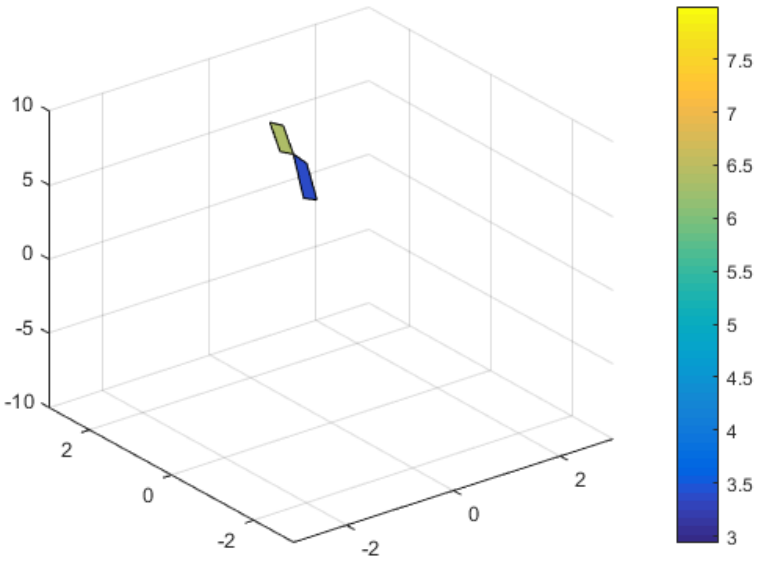

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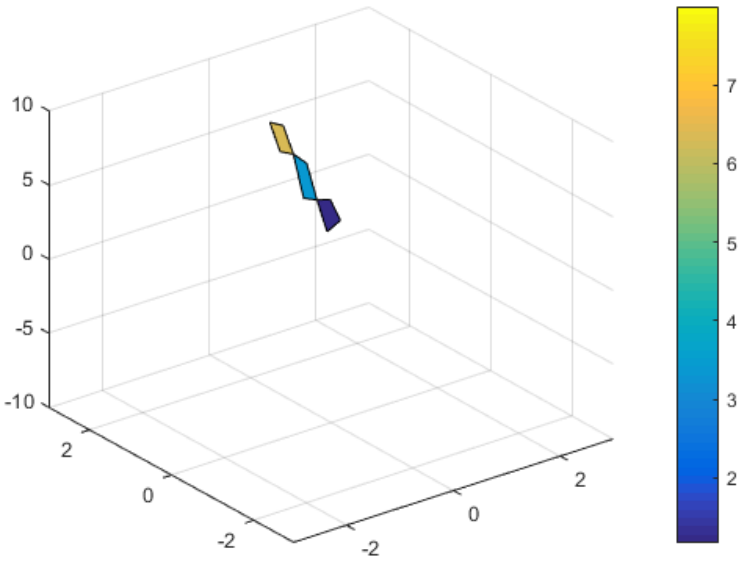

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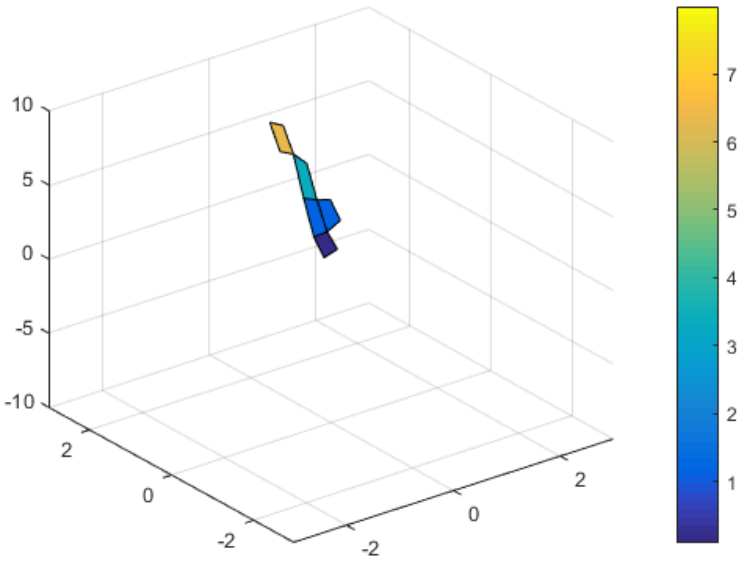

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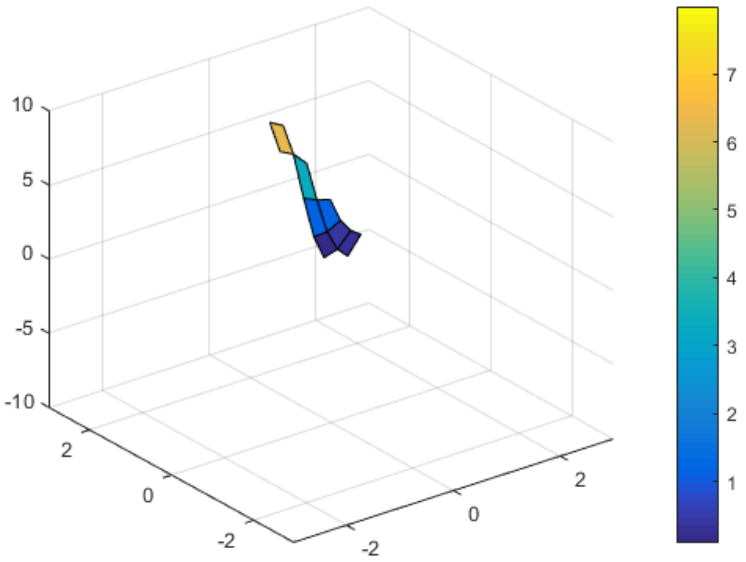

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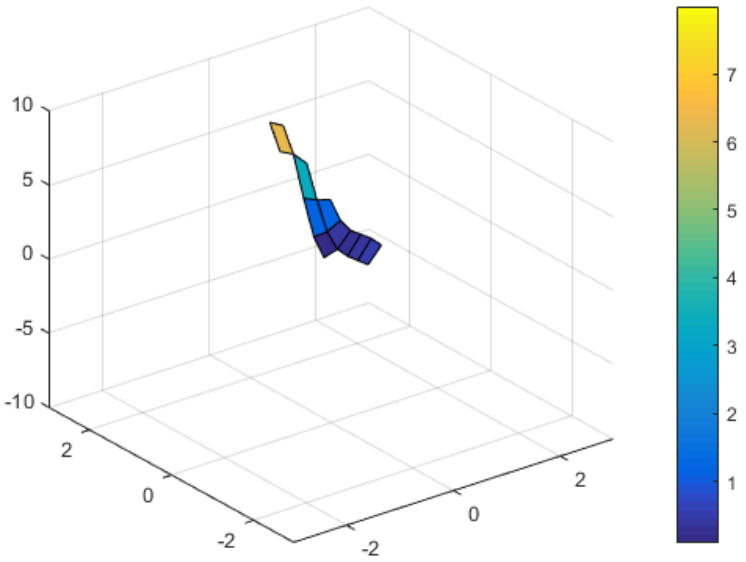

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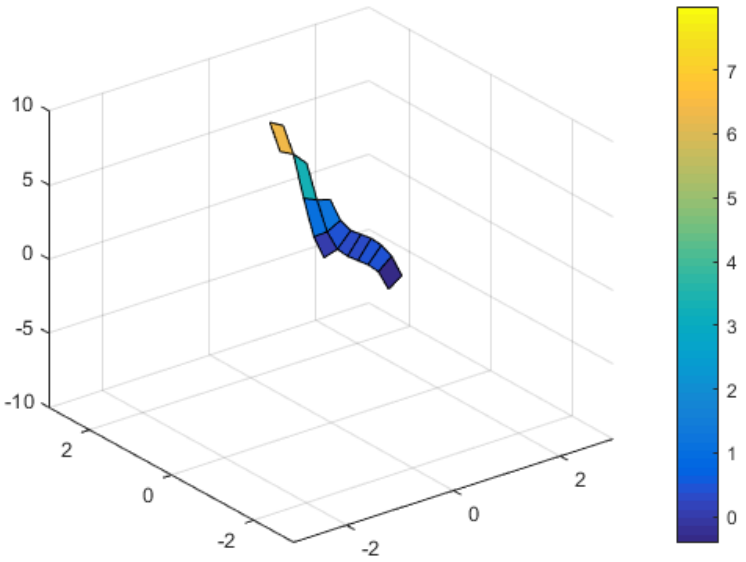

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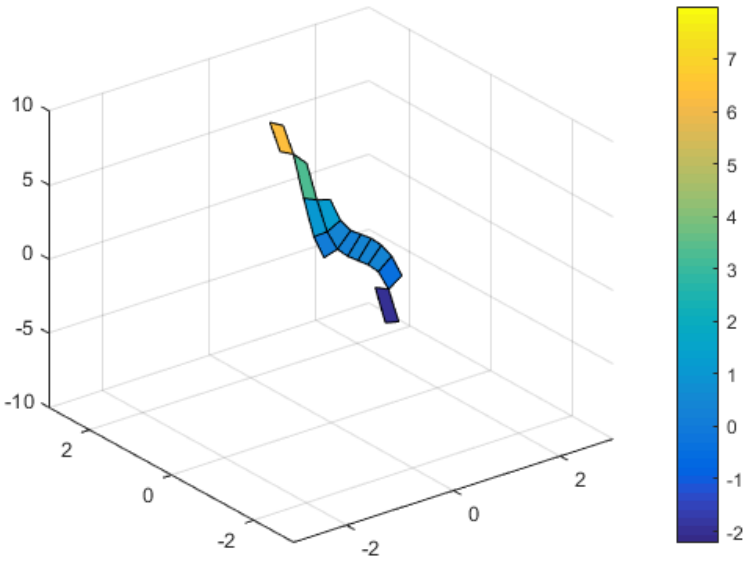

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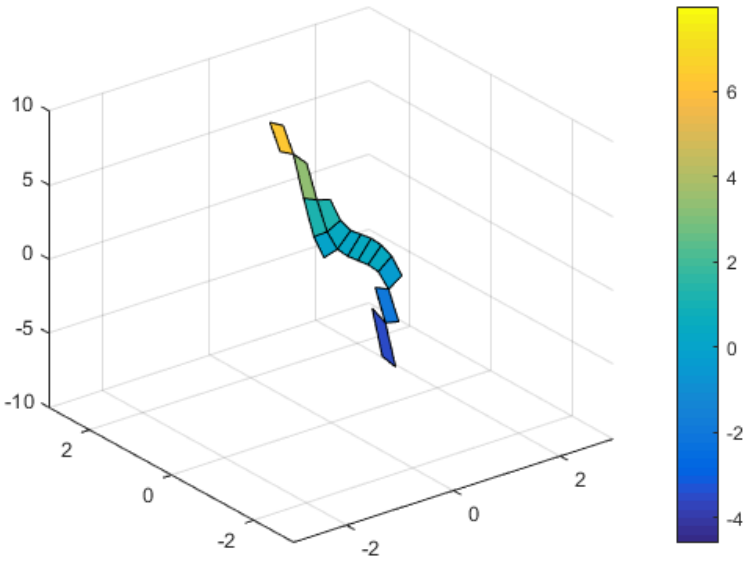

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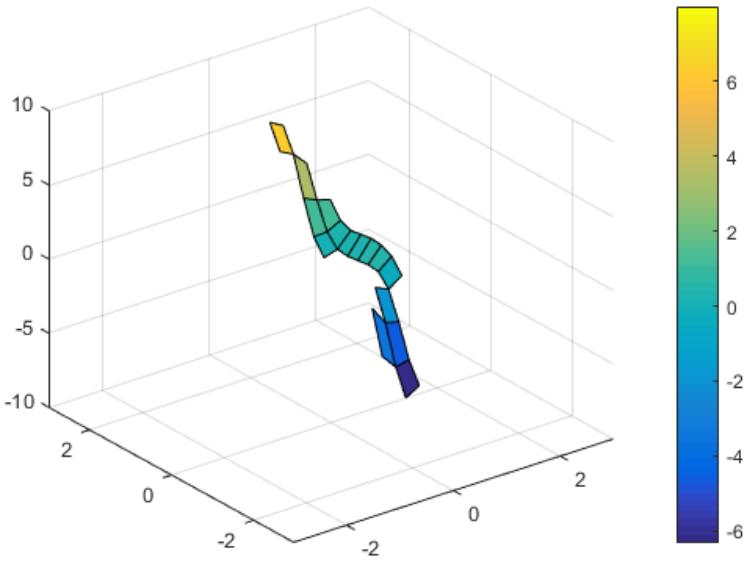

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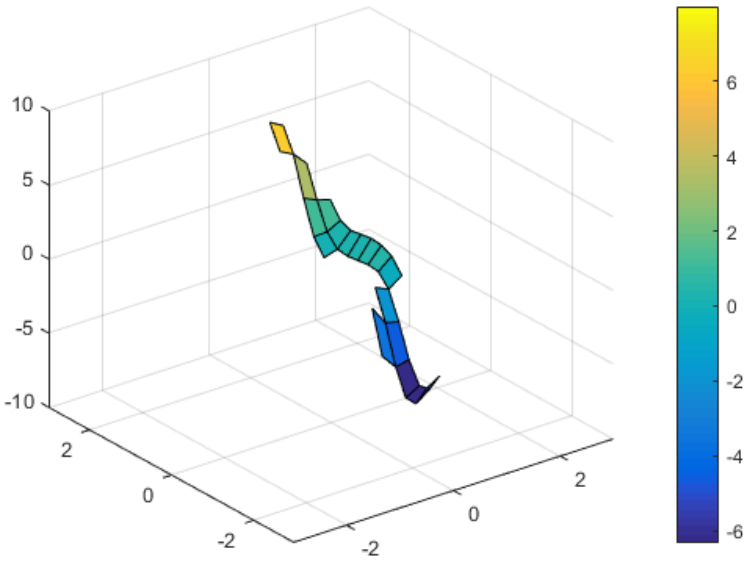

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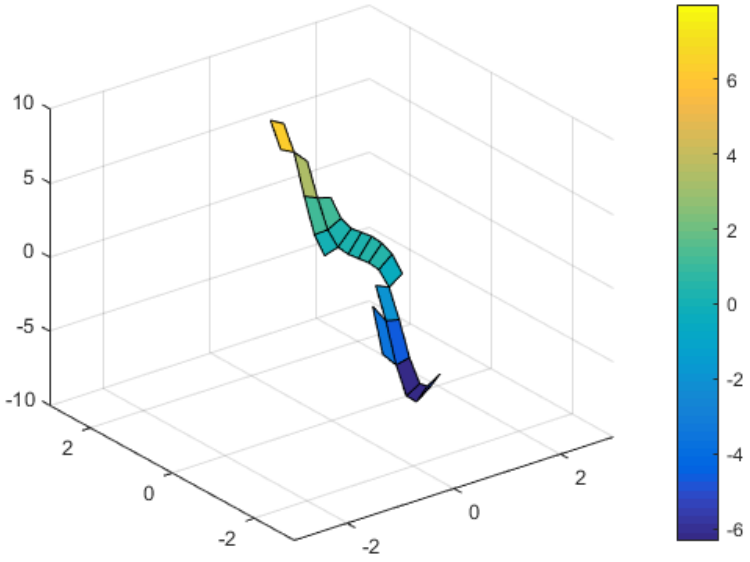

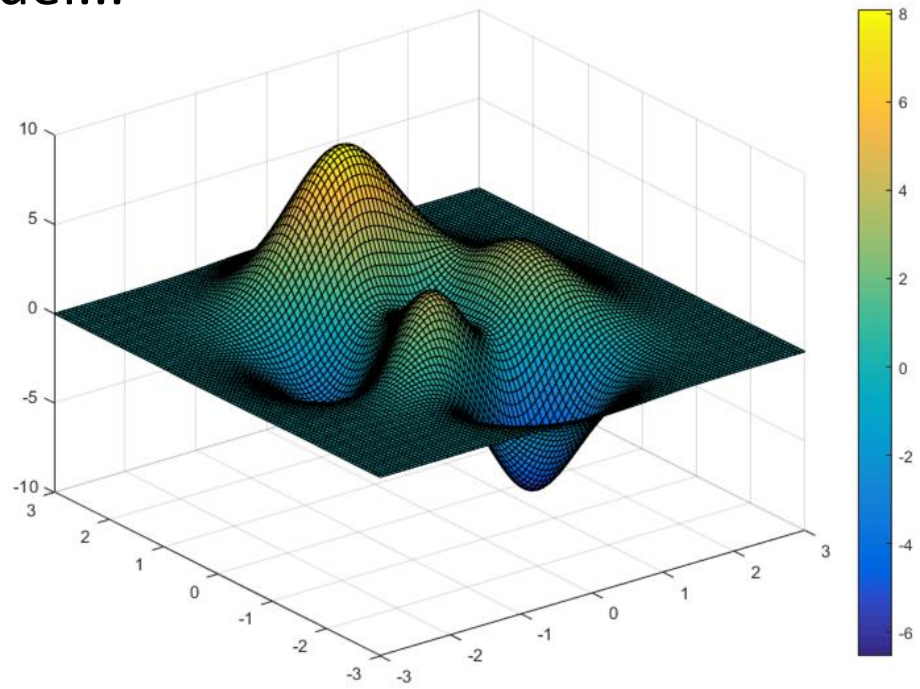

We found the minimum with 10 computations vs 1000+ from direct enumeration

Parallel Hybrid Design and Intro to Optimization

- Next time:
  - More parallel HEV design
  - Series/Parallel or Power-Split HEV design
  - Optimization continued….